Python as a Language

Interpretation, History & Evolution

Programming Fundamentals Team

SETU

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Outline

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1. What is Python?

- High-level, dynamically-typed, interpreted programming language
- Created by Guido van Rossum in 1991
- Designed to be easy to read and write
- Versatile: web, data science, automation, AI, etc.
- Open-source and free
- Named after “Monty Python’s Flying Circus”

Outline

2.1 Interpreted Language

The Execution Process:

1. **Writing the Code** — Create a `.py` file
2. **Parsing** — Interpreter reads your code
3. **Compilation to Bytecode** — Source code converted to bytecode
4. **Bytecode Execution** — Python Virtual Machine executes bytecode

```
Your Python Code (.py)
        ↓
Python Interpreter
        ↓
Bytecode (.pyc)
        ↓
Python Virtual Machine (PVM)
        ↓
Machine Code (CPU)
```

2.2 Interpreted vs Compiled Languages

Compiled (C, C++)

- Fast execution
- Compiled once
- Errors at compile time
- Platform-specific
- Slower development

Interpreted (Python)

- Slower execution
- Compiled at runtime
- Errors at runtime
- Cross-platform
- Faster development

2.3 Python's Virtual Machine (PVM)

The **Python Virtual Machine** is an abstract computing machine that executes Python bytecode.

It's the layer between your code and the actual computer hardware.

This design enables:

- **Development Speed** — No compilation step needed
- **Cross-Platform** — Same code runs everywhere
- **Portability** — Windows, Mac, Linux without changes

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3.1 Historical Timeline

1989 ⟶ Guido van Rossum creates Python at CWI (Amsterdam) starts development

1991 ⟶ Python 0.9 released

1994 ⟶ Python 1.0 (first stable)

2000 ⟶ Python 2.0 (industry standard)

2008 ⟶ Python 3.0 (breaking change)

2010s ⟶ Python 2 vs 3 war

2020 ⟶ Python 2 officially ends

2024 ⟶ Python 3.12+ (modern, fast)

3.2 Python Implementations

Different implementations of the Python language:

- **CPython** — Most common (written in C)
- **PyPy** — Faster with JIT compilation
- **Jython** — Runs on Java Virtual Machine
- **IronPython** — Runs on .NET

3.3 Why Python Exploded in Popularity

1. **Data Science Boom** — NumPy, Pandas, Scikit-learn
2. **AI/Machine Learning** — TensorFlow, PyTorch, default choice
3. **Simple Syntax** — Perfect for teaching
4. **Web Development** — Django, Flask frameworks
5. **Automation** — DevOps and scripting
6. **Community** — Stack Overflow, GitHub, Reddit

3.4 Adoption Milestones

- **1994-2000** — System administration and scripting
- **2006-2010** — Data science libraries emerge
- **2010-2015** — Machine learning revolution
- **2015+** — 1 language for AI/ML
- **2020+** — Surpasses Java in popularity

3.5 Current State (2024)

Python is now:

- **1–2 most popular language** (competing with JavaScript)
- **Default for data science, AI, and ML**
- **Growing in web development** (FastAPI, etc.)
- **Essential in education** — taught worldwide
- **Industry standard** for automation and DevOps

Outline

4. Guido van Rossum's Philosophy

The "Zen of Python" (PEP 20) summarizes the language:

Beautiful is better than ugly

Explicit is better than implicit

Simple is better than complex

Readability counts

This philosophy has guided Python for 30+ years and explains its success.

Thanks for Watching - Any questions?